

IS 670 — Lecture 7

Numeric Prediction — Regression Methods

Complete Tutor Document • From First Principles • Lecture 7 + Lab 7

How to read this document

Each section follows the same structure: plain-English definition, problem it solves, inputs/outputs, how it works, why it works, when to use (and not), strengths/weaknesses, assumptions, formulas, visuals, and code from Lab 7. Work through it slowly; do not skip the worked examples.

1. Big Picture — What This Lecture Is About

Lecture 7 is the start of numeric prediction. Up to Lecture 6 you learned how to predict a category (cat vs dog, good-buy vs bad-buy). Starting in Lecture 7, the target variable is a number on a continuous scale — for example, the cost of medical insurance, the price of a house, or the number of units a store will sell next week.

Why business cares

- Pricing — what should we charge this customer? (insurance premium, Uber fare)
- Forecasting — how many units will we sell? (demand planning)
- Risk — what is the expected claim size for this policy?
- Valuation — what is this house or car worth?

What changes from classification

- The output y is a real number, not a class label.
- Accuracy / precision / recall no longer apply. We use MAE and RMSE instead.
- New metric: R^2 — how much of the variation in y the model explains.
- Residual = $y_{\text{true}} - y_{\text{pred}}$ (the error on one row). Errors are now numeric, not just right/wrong.

The core question for every regression model

Given this new row of features, what is our best single number prediction for y , and how wrong is the model on average?

2. Concepts Covered in Lecture 7

1. Supervised learning — regression vs classification
2. Simple Linear Regression (one predictor)
3. Multiple Linear Regression (many predictors)
4. Ordinary Least Squares (OLS) — how the line is fit
5. Coefficients (β), intercept, p-values, R^2 , Adjusted R^2
6. Dummy coding / binary indicators for categorical predictors
7. Train / Test split
8. Residuals
9. Assumptions of linear regression
10. Evaluation metrics — MAE and RMSE
11. Non-linear terms (polynomial) and interaction effects
12. Regression Trees (split on Squared Error Reduction)
13. KNN Regression

3. In-Depth — Each Concept from Zero

3.1 Simple Linear Regression

Plain English

Draw the best straight line through a scatter of (x, y) points. The line lets you predict y for any new x.

Problem it solves

You want to predict a number y from one numeric predictor x.

Input / Output

- Input: one numeric predictor x (e.g., age).
- Output: one numeric prediction $\hat{y}$ (e.g., insurance charge).

How it works (step-by-step)

14. Assume the relationship is a straight line: $\hat{y} = \beta_0 + \beta_1 \cdot x$.
15. For every candidate line, compute the residual for each row: $e_i = y_i - \hat{y}_i$.
16. Compute the Sum of Squared Errors: $SSE = \sum e_i^2$.
17. Pick β_0 and β_1 that minimize SSE. This is what OLS does — it has a closed-form answer, no iteration needed.
18. To predict a new point, plug its x into the fitted line.

Why it works (intuition)

Squaring the errors penalizes large mistakes more than small ones, so the line is pulled toward the cloud of points as a whole. Using squared (not absolute) errors also makes the math have a clean closed-form solution.

When to use / When NOT to use

- Use when the relationship between x and y looks roughly linear on a scatter plot.
- Don't use when the relationship is clearly curved (U-shaped, saturating, etc.) — the line will systematically under/over-predict.

Strengths

- Interpretable: one unit increase in x changes $\hat{y}$ by β_1 .
- Fast to fit, fast to predict.
- Gives p-values and confidence intervals (if you use statsmodels).

Weaknesses

- Only captures straight-line relationships.
- Sensitive to outliers (the squaring amplifies big errors).
- Requires assumptions (see 3.9) to trust the p-values.

3.2 Multiple Linear Regression

Same idea but with many predictors:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$$

Each β_j says: "if x_j goes up by 1 and everything else is held constant, $\hat{y}$ changes by β_j ." That "holding everything else constant" part is the single most important idea to understand about multiple regression.

Example from Lab 7

Predict medical charges from age, bmi, children, sex, smoker, region. Each β tells the per-unit effect of one predictor while controlling for the others. In the lab, the smoker coefficient is very large and positive — smokers pay far more, even after accounting for age and bmi.

3.3 Ordinary Least Squares (OLS)

OLS is the fitting procedure: find the β values that minimize $\sum(y_i - \hat{y}_i)^2$. For simple linear regression:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \cdot \bar{x}\end{aligned}$$

Symbol legend:

- x_i, y_i — value of predictor / target for row i
- $\bar{x}, \bar{y}$ — mean of x , mean of y across the training data
- $\hat{\beta}_0$ — estimated intercept (predicted y when $x = 0$)
- $\hat{\beta}_1$ — estimated slope (change in $\hat{y}$ per +1 in x)

3.4 Reading an OLS Output — β , p-values, R^2 , Adjusted R^2

- Coefficient (β) — direction and size of the effect. Sign tells the direction.
- Std error — uncertainty around the coefficient.
- t-stat = β / SE . Big $|t|$ means the coefficient is likely non-zero.
- p-value — probability of seeing this β (or bigger) if the true effect were zero. $p < 0.05$ = "statistically significant".
- R^2 — fraction of the variance of y explained by the model. 0 = nothing, 1 = perfect.
- Adjusted R^2 — R^2 adjusted for the number of predictors, so you cannot inflate it by throwing in junk features.

Exam trap

A high R^2 does not mean the model is right. R^2 only tells you how well the line fits THIS training data. Always evaluate MAE / RMSE on the TEST set to check generalization.

3.5 Dummy Coding / Binary Indicators

Regression needs numbers. Categorical columns like "sex" (male/female) or "region" (NE, NW, SE, SW) are converted to 0/1 columns.

Rule: for a category with k levels, create $k - 1$ dummies. One level is the "reference" and is represented by all zeros. This avoids the dummy-variable trap (perfect multicollinearity).

```
pd.get_dummies(df, drop_first=True, dtype=int)
```

Interpreting a dummy coefficient

Example: smoker_yes has $\beta = +23,800$. Interpretation: "being a smoker adds \$23,800 to predicted charges compared to a non-smoker, everything else equal."

3.6 Train / Test Split

Fit the model on the training subset, evaluate on the test subset (the model has never seen it). This is how we measure generalization, not memorization.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

3.7 Residuals

$e_i = y_i - \hat{y}_i$. Residuals are the raw errors on each row. We use them to build MAE, RMSE, and to diagnose whether the model assumptions hold (see 3.9).

3.8 Evaluation Metrics — MAE and RMSE

Metric	Formula	Meaning	Units
MAE	$(1/n) \sum y_i - \hat{y}_i $	Average absolute error. Robust to outliers.	Same as y (e.g., dollars)
MSE	$(1/n) \sum (y_i - \hat{y}_i)^2$	Average squared error. Penalizes big errors more.	y^2
RMSE	$\sqrt{\text{MSE}}$	Like MAE but punishes large mistakes more.	Same as y
R^2	$1 - \text{SSR}/\text{SST}$	Fraction of variance explained.	unitless (0 to 1)

Lower MAE / RMSE = better. Higher R^2 = better.

MAE vs RMSE — when does the difference matter?

RMSE is always $\geq$ MAE. A much larger RMSE than MAE tells you there are a few very large errors (tail risk). If those rare big misses matter for the business (e.g., mis-pricing a whale customer), minimize RMSE.

3.9 Assumptions of Linear Regression

Assumption	What it means	How to check
Linearity	The true relationship between x and y is a straight line.	Scatter plot; residuals vs fitted plot
Independence	Each observation is independent of the others.	Domain knowledge; time-series = usually violated
Homoscedasticity	Error variance is constant across fitted values.	Residuals vs fitted — no funnel shape
No perfect multicollinearity	No predictor is a perfect linear combination of others.	Correlation matrix; VIF
Zero conditional mean	$E[\text{error} x] = 0$ (no systematic bias).	Mean of residuals ≈ 0 for each x bin
Normality of errors	Residuals are approximately normal.	Histogram of residuals; Q-Q plot

3.10 Non-Linear Terms and Interaction Effects

Polynomial (quadratic, cubic)

If y bends with x, add x^2 as an extra predictor. Example: medical charges vs age might curve, so include both age and age^2 (in Lab 7: "age2").

Interaction

The effect of one predictor depends on another. Example: being a smoker adds a lot — but being a smoker with high bmi adds even more. Model it as a new column $\text{smoker} \times \text{bmi} \geq 30$.

```
df.loc[:, 'age2'] = df['age'] ** 2
df.loc[:, 'bmi30_smoker'] = ((df['bmi'] >= 30) & (df['smoker'] == 'yes')).astype(int)
```

3.11 Regression Trees

Same decision tree idea as classification, but the leaf prediction is the MEAN of the y values that fall into that leaf, and the split criterion is based on reducing variance (equivalently, squared error).

Split criterion — SDR (Standard Deviation Reduction) / Squared Error Reduction

$$\text{SDR} = \text{sd}(T) - \sum (|T_i| / |T|) \cdot \text{sd}(T_i)$$

T = node before split. T_i = children. We choose the split that reduces the standard deviation of y the most inside the children.

Strengths

- Captures non-linear patterns automatically.
- Handles mixed numeric + categorical features.
- Very interpretable — you can draw the tree.

Weaknesses

- Unstable — small data change can flip the tree.
- Easy to overfit if max_depth is too large.
- Predictions are piecewise constant (jumpy), not smooth.

3.12 K-Nearest Neighbors Regression

To predict $\hat{y}$ for a new row: find the K closest training rows by Euclidean distance, then AVERAGE their y values.

Important: features must be on the same scale (use MinMaxScaler) or the features with large units dominate the distance.

- K too small (e.g., K=1) → overfits, noisy predictions.
- K too large → predictions shrink toward the global mean, underfits.
- No model is "trained" — KNN is a lazy learner; all the work happens at prediction time.

4. Formulas Cheat Sheet

Concept	Formula	Symbols
Simple linear model	$\hat{y} = \beta_0 + \beta_1 x$	β_0 intercept, β_1 slope
Multiple linear model	$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$	one β per predictor
OLS estimator (simple)	$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$; $\beta_0 = \bar{y} - \beta_1 \bar{x}$	means $\bar{x}$, $\bar{y}$
Residual	$e_i = y_i - \hat{y}_i$	row-level error
MAE	$(1/n) \sum y_i - \hat{y}_i $	avg absolute error
MSE	$(1/n) \sum (y_i - \hat{y}_i)^2$	avg squared error
RMSE	$\sqrt{\text{MSE}}$	same units as y
R^2	$1 - \text{SSR}/\text{SST} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$	0–1, higher better
Adjusted R^2	$1 - (1 - R^2)(n-1)/(n-k-1)$	k predictors, n rows
Euclidean distance	$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$	for KNN
Tree split (SDR)	$\text{sd}(T) - \sum (T_i / T) \cdot \text{sd}(T_i)$	pick split with max SDR

5. Key Visuals — How to Read Them

Visual	What it shows	How to interpret
Scatter of y vs x with fitted line	Raw data and the regression line	Line going up = positive relationship; points close to line = low error
Residual plot (residual vs fitted)	Errors vs predictions	Should look like a random cloud around 0; any shape (funnel, curve) = assumption violated
OLS summary table (statsmodels)	β , SE, t, p, R^2 , Adj R^2	Check sign of β , check $p < 0.05$ for significance, check R^2 for fit
Regression tree plot	Splits and leaf means	Each leaf shows the mean y prediction for rows that reach it
MAE / RMSE bar comparison	Errors for LinReg vs Tree vs KNN	Lower = better; compare on the TEST set, not train

6. Lab 7 — Code Line by Line

Dataset: insurance.csv (age, sex, bmi, children, smoker, region, charges). Goal: predict charges.

6.1 Load and inspect

```
import pandas as pd
import numpy as np
import seaborn as sns
df = pd.read_csv('insurance.csv')
df.head()
df.info()
df.describe()
```

- read_csv — load the file into a DataFrame.
- head — peek at first 5 rows (sanity check columns look right).
- info — dtypes and missing counts.
- describe — numeric summary (mean, std, min, max, quartiles).

6.2 Fix column types and create dummies

```
df['sex'] = df['sex'].astype('category')
df['smoker'] = df['smoker'].astype('category')
df['region'] = df['region'].astype('category')
df = pd.get_dummies(df, drop_first=True, dtype=int)
```

- astype('category') — tell pandas these are discrete labels, not free-text strings.
- get_dummies — creates 0/1 columns. drop_first=True keeps k-1 dummies to avoid the dummy trap.
- dtype=int — keeps the new columns as integers (readable 0/1).

6.3 Define X, y and split train/test

```
from sklearn.model_selection import train_test_split
y = df['charges']
X = df.drop(columns=['charges'])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)
```

random_state=42 makes the split reproducible. test_size=0.3 keeps 30% of rows unseen for evaluation.

6.4 Linear regression (sklearn)

```
from sklearn.linear_model import LinearRegression
lin = LinearRegression().fit(X_train, y_train)
pred_lin = lin.predict(X_test)
print('Intercept:', lin.intercept_)
print('Coefficients:', dict(zip(X.columns, lin.coef_)))
```

Interpretation: the smoker_yes coefficient will be very large and positive (~+\$23k), confirming smoking is the dominant cost driver.

6.5 OLS summary (statsmodels) — for p-values and R²

```
import statsmodels.api as sm
X_train_sm = sm.add_constant(X_train)
```

```
ols = sm.OLS(y_train, X_train_sm).fit()
print(ols.summary())
```

add_constant — adds the column of 1s so the intercept is estimated. In the summary look at:

- coef column — sign and magnitude of each β
- $P > |t|$ — p-values; flag predictors with $p < 0.05$ as significant
- R-squared and Adj. R-squared — overall model fit

6.6 MAE and RMSE on the test set

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
mae = mean_absolute_error(y_test, pred_lin)
rmse = mean_squared_error(y_test, pred_lin) ** 0.5
print('MAE:', mae, 'RMSE:', rmse)
```

6.7 Improve the model — age² and smoker×obesity

```
df.loc[:, 'age2'] = df['age'] ** 2
df.loc[:, 'bmi30_smoker'] = ((df['bmi'] >= 30) & (df['smoker_yes'] == 1)).astype(int)
```

Re-fit the model with these two extra columns — R^2 jumps, MAE and RMSE drop. The lab demonstrates that simple engineered features can beat a fancier model.

6.8 Regression tree

```
from sklearn.tree import DecisionTreeRegressor, plot_tree
tree = DecisionTreeRegressor(max_depth=3, random_state=1).fit(X_train, y_train)
pred_tree = tree.predict(X_test)
plot_tree(tree, feature_names=X.columns, filled=True)
```

Inspect the plotted tree — the first split will usually be on smoker because it explains the most variance.

6.9 KNN regression

```
from sklearn.neighbors import KNeighborsRegressor
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler().fit(X_train)
X_train_s = scaler.transform(X_train)
X_test_s = scaler.transform(X_test)
knn = KNeighborsRegressor(n_neighbors=5).fit(X_train_s, y_train)
pred_knn = knn.predict(X_test_s)
```

Scaling rule

Fit MinMaxScaler on the training data only, then transform both train and test. Never fit on the test set — that is data leakage.

7. Beginner Worked Example (by hand)

Suppose we have 4 training rows of age (x) and charges in \$ (y):

age (x)	charges (y)
20	5000

30	8000
40	11000
50	14000

Means: $\bar{x} = 35$, $\bar{y} = 9500$.

Numerator of β_1 : $\sum (x_i - \bar{x})(y_i - \bar{y}) = (-15)(-4500) + (-5)(-1500) + (5)(1500) + (15)(4500) = 67500 + 7500 + 7500 + 67500 = 150000$.

Denominator: $\sum (x_i - \bar{x})^2 = 225 + 25 + 25 + 225 = 500$.

$\beta_1 = 150000 / 500 = 300$. $\beta_0 = 9500 - 300 \cdot 35 = -1000$.

So $\hat{y} = -1000 + 300 \cdot \text{age}$. For a 45-year-old: $\hat{y} = -1000 + 300 \cdot 45 = \$12,500$.

8. Exam-Style Interpretation Examples

Q. "Given the OLS summary below, the coefficient on smoker_yes is +23,800 with $p < 0.001$. Interpret."

A. Holding age, bmi, children, sex, and region constant, being a smoker is associated with \$23,800 higher predicted charges than being a non-smoker. The p-value < 0.001 means this effect is statistically significant — very unlikely to be due to random chance.

Q. "Model A: MAE=4200, RMSE=6800. Model B: MAE=4100, RMSE=8900. Which is better?"

A. It depends on business cost. Model B has slightly lower average error (MAE) but much higher RMSE, meaning it occasionally makes very large mistakes. If large individual mis-pricings are costly, prefer Model A.

Q. " R^2 on training = 0.87, R^2 on test = 0.61. What does this tell you?"

A. Overfitting. The model memorizes the training set but does not generalize. Reduce model complexity (fewer predictors, smaller tree depth, larger K).

9. Concluding Summaries

9.1 Simple Summary (plain English)

Lecture 7 is about predicting a number. Linear regression draws the best straight line; OLS does the math that finds it. R^2 tells you how well the line fits, p-values tell you which predictors matter, and MAE/RMSE tell you how wrong you are on new data. If the real pattern is curvy, add squared or interaction terms, or switch to a regression tree or KNN.

9.2 Technical Summary

Regression assumes $\hat{y} = X\beta$ with OLS estimator $\hat{\beta} = (X^T X)^{-1} X^T y$ minimizing $\sum (y_i - \hat{y}_i)^2$. Inference requires Gauss-Markov assumptions (linearity, independence, homoscedasticity, no perfect multicollinearity, zero conditional mean) plus normality for exact t-tests. Diagnostics: residual plot, Q-Q plot, VIF. Alternatives: regression trees partition X into axis-aligned regions using SDR as split criterion; KNN regression averages the K nearest neighbors (distance-based, requires scaling).

9.3 Key Formulas Recap

- $\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$
- $MAE = (1/n) \sum |y_i - \hat{y}_i|$ • $RMSE = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2}$
- $R^2 = 1 - SSR/SST$
- $SDR = sd(T) - \sum (|T_i|/|T|) \cdot sd(T_i)$
- Euclidean $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$

9.4 Key Definitions Recap

Term	Meaning
Regression	Predict a numeric target.
OLS	Method that minimizes $\sum$ squared residuals.
Coefficient (β)	Effect on $\hat{y}$ of +1 unit of a predictor, others held constant.
Residual	$y_{\text{true}} - y_{\text{pred}}$ for one row.
R^2	Variance of y explained by the model (0 to 1).
Adjusted R^2	R^2 penalized for extra predictors.
MAE	Average absolute error.
RMSE	Square root of the average squared error.
Dummy	0/1 column encoding a categorical level.
Interaction	Product of two predictors to model combined effect.
Regression tree	Tree whose leaves predict the mean y of rows in that leaf.
KNN regression	Predict the average y of the K nearest training rows.

9.5 Common Mistakes

- Evaluating on the training set (R^2 looks great, real performance is worse).
- Forgetting `drop_first=True` and creating perfect multicollinearity.
- Leaving categorical columns as strings (model errors out).
- Using accuracy for a regression problem.
- Fitting `MinMaxScaler` on test data (leakage).
- Reading β as causal. It is correlational — controls for everything that is in the model, not everything in the world.

9.6 Common Exam Traps

- High $R^2 \neq$ good model. Check test RMSE.
- Significance ($p < 0.05$) $\neq$ practically important. A tiny β can be "significant" with enough data.
- A negative β on smoker_yes would be suspicious — always sanity-check sign.
- Lower MAE does not imply lower RMSE — they can disagree.
- Tree depth 1 is a stump; don't call it a "deep" tree.

10. Practice Questions (with answers)

Q1. Given simple linear regression on 4 points with $\bar{x}=35$, $\bar{y}=9500$, $\Sigma(x_i-\bar{x})(y_i-\bar{y})=150000$, $\Sigma(x_i-\bar{x})^2=500$, compute β_0 and β_1 .

Answer: $\beta_1 = 150000/500 = 300$. $\beta_0 = 9500 - 300 \cdot 35 = -1000$.

Q2. You run OLS on insurance data and get $R^2 = 0.75$ on train and 0.74 on test. Is this overfitting?

Answer: No — train and test R^2 are essentially equal, so the model generalizes. It would be overfitting if train were much higher than test.

Q3. A dummy variable region_southwest has $\beta = -800$, $p = 0.42$. Interpret.

Answer: Compared to the reference region, being in the southwest is associated with \$800 lower predicted charges — but $p = 0.42$ means this effect is not statistically distinguishable from zero. Do not rely on it.

Q4. Why is $RMSE \geq MAE$ always true?

Answer: Squaring the errors before averaging gives extra weight to large errors, then the square root is taken. Mathematically, the quadratic mean is always $\geq$ the arithmetic mean of the absolute values (Jensen's inequality).

Q5. You add 5 random noise columns to your regression. R^2 goes up. Does that mean the model is better?

Answer: No — R^2 never decreases when you add predictors, even garbage. Use Adjusted R^2 or test-set RMSE to judge real improvement.

11. Practice Questions (no answers — try yourself)

Q1. In Lab 7 the coefficient on age2 turns out positive. What business story does this tell about the age–charges relationship?

Q2. The OLS summary shows children with $\beta = +475$, $p = 0.02$. Translate this into a sentence a non-technical manager would understand.

Q3. A regression tree with `max_depth=10` has training MAE = 900 and test MAE = 4200. What went wrong and what do you do about it?

Q4. You use KNN regression without scaling. Charges has range [1k, 60k] and age has range [18, 64]. Which feature dominates the distance, and why is that bad?

Q5. Name one situation where a regression tree is preferable to multiple linear regression, and one where linear regression is preferable to a tree.

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